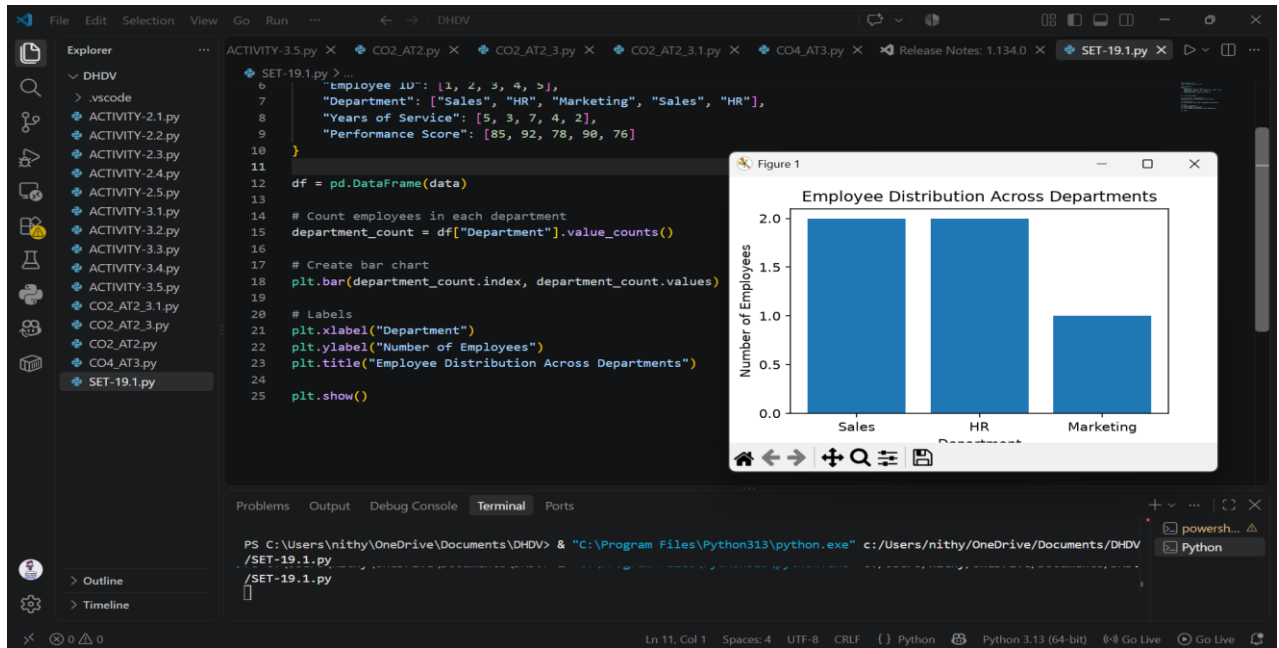


SET-19

QUESTION-1

Aim

To create a bar chart showing the number of employees in each department.



CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

data = {

    "Employee ID": [1, 2, 3, 4, 5],

    "Department": ["Sales", "HR", "Marketing", "Sales", "HR"],

    "Years of Service": [5, 3, 7, 4, 2],

    "Performance Score": [85, 92, 78, 90, 76]

}

df = pd.DataFrame(data)

department_count = df["Department"].value_counts()
```

```
plt.bar(department_count.index, department_count.values)
plt.xlabel("Department")
plt.ylabel("Number of Employees")
plt.title("Employee Distribution Across Departments")
plt.show()
```

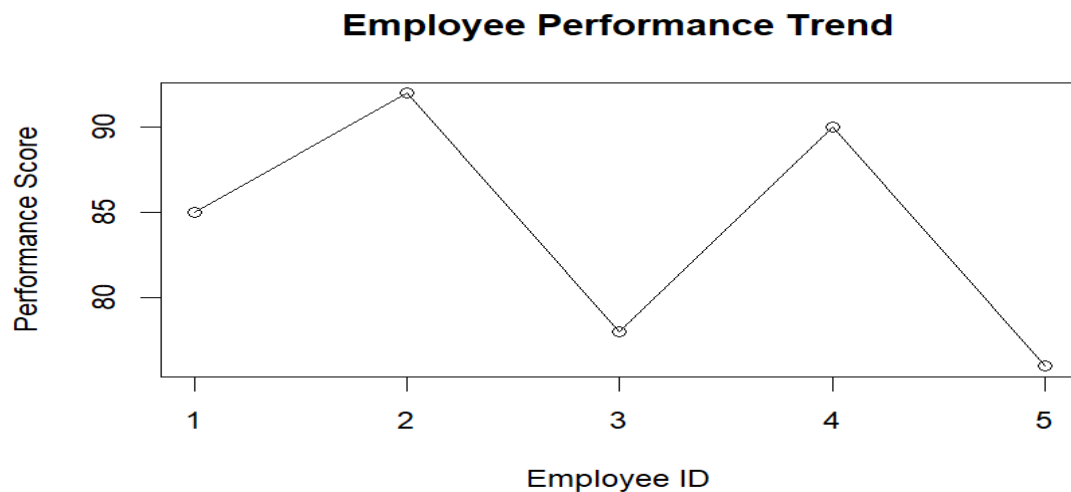
Result

Thus, a bar chart was created successfully to visualize the distribution of employees across different departments.

QUESTION -2

Aim

To generate a line chart showing the performance scores of employees.



CODE:

```
employee_id <- c(1, 2, 3, 4, 5)
performance <- c(85, 92, 78, 90, 76)
plot(employee_id, performance,
      type = "o",
      xlab = "Employee ID",
      ylab = "Performance Score",
      main = "Employee Performance Trend")
```

Result

Thus, a line chart was successfully created to visualize the employee performance trend.

Note: Since the dataset does not contain an actual time column, Employee ID is used as the sequence for visualization

QUESTION -4

Aim

To develop and display a table containing the performance details of each employee.

```
/SET-19.1.py
PS C:\Users\nithy\OneDrive\Documents\DHDV> & "C:\Program Files\Python313\python.exe" c:/Users/nithy/OneDrive/Documents/DHDV/EXP-19.4.py
Employee Performance Data
Employee ID Department Years of Service Performance Score
0 1 Sales 5 85
1 2 HR 3 92
2 3 Marketing 7 78
3 4 Sales 4 90
4 5 HR 2 76
PS C:\Users\nithy\OneDrive\Documents\DHDV>
```

CODE:

```
import pandas as pd

data = {
    "Employee ID": [1, 2, 3, 4, 5],
    "Department": ["Sales", "HR", "Marketing", "Sales", "HR"],
    "Years of Service": [5, 3, 7, 4, 2],
    "Performance Score": [85, 92, 78, 90, 76]
}

df = pd.DataFrame(data)

print("Employee Performance Data")

print(df)
```